

'Rente mit 63'

Labour Force Participation among Eligible Older Workers in Germany

Reform: July 1, 2014

Regression Results

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SPS Pareto Seminar · Research Projects in Empirical Economics · 2026

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Model 1 — Control Group: 35–44 Contribution Years



Regression Results

DiD Coefficient: **-0.170**

Std. Error: **0.053**

p-value: **0.001 *****

Sample Sizes

N Treatment: 572

N Control: 5,981

Ratio (C:T): **10.5 : 1**

Parallel Trends Test

p = 0.144 ✓ Assumption satisfied

✓ Pros

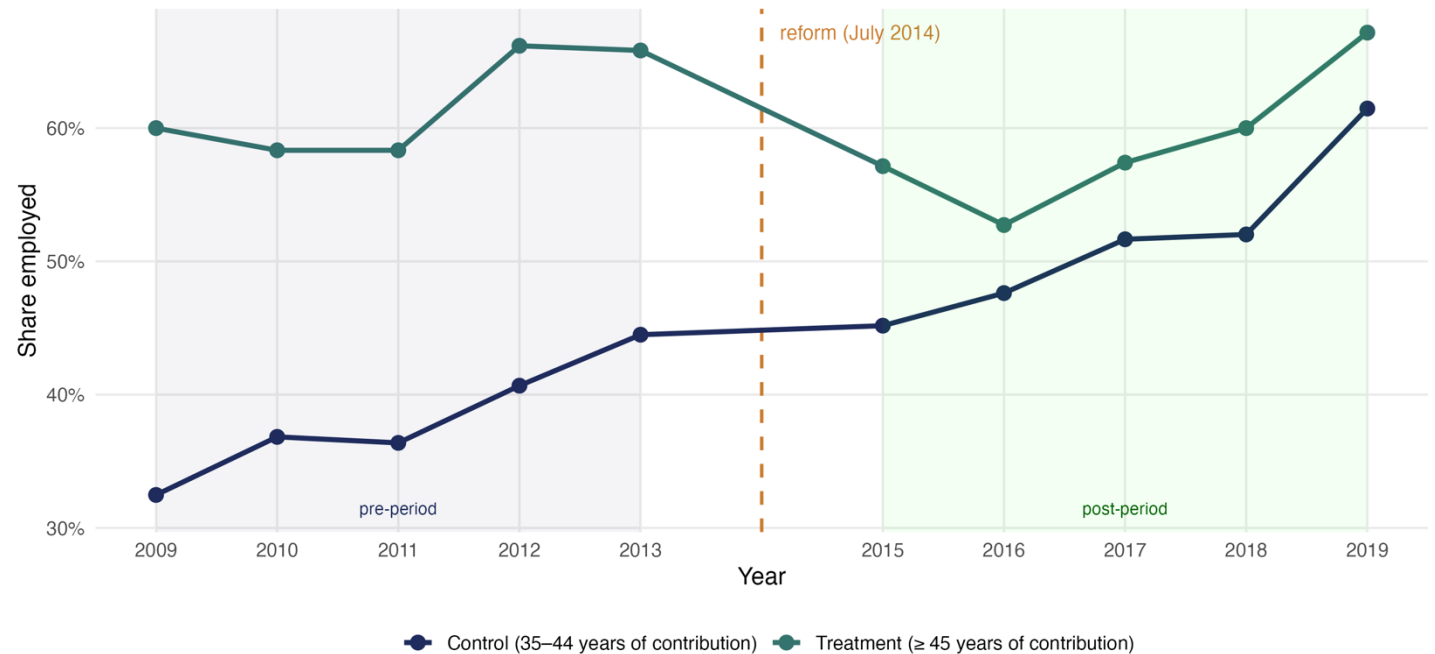
Large control group
High statistical power
Wide coverage

✗ Cons

Control too heterogeneous
Ratio 10.5:1 imbalanced

Employed: treatment- vs. controlgroup (2009–2019)

Control: 35–44 contribution years | 2014 excluded | Parallel Trends: p = 0.144



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Model 2 — Control Group: 40–44 Contribution Years



Regression Results

DiD Coefficient: **-0.137**

Std. Error: **0.055**

p-value: **0.013 ***

Sample Sizes

N Treatment: 572

N Control: 2,866

Ratio (C:T): **5.0 : 1**

Parallel Trends Test

p = 0.108 ✓ Assumption satisfied

✓ Pros

PT assumption holds

Closer to treatment group

Good sample size

✗ Cons

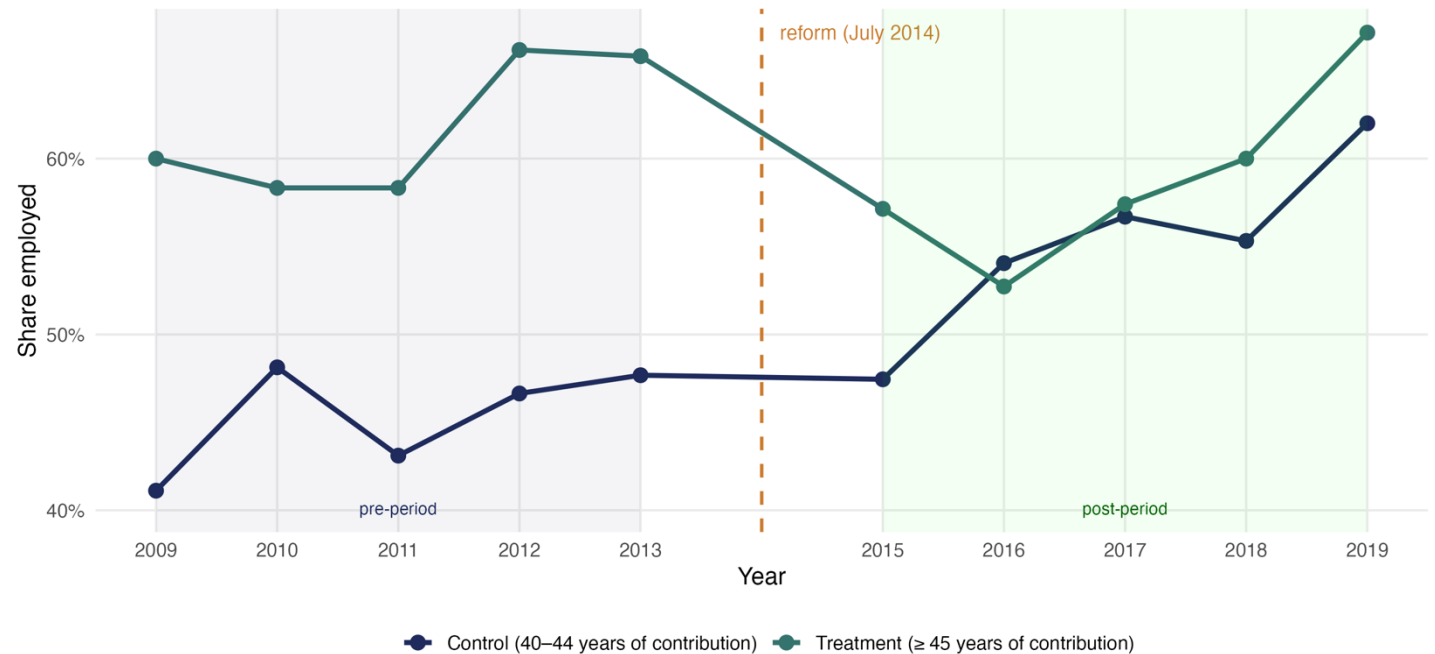
Ratio still 5.0:1

Control range still broad

Some residual heterogeneity

Employed: treatment- vs. controlgroup (2009–2019)

Control: 40–44 contribution years | 2014 excluded | Parallel Trends: p = 0.108



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Model 3 — Control Group: 41–44 Contribution Years



Regression Results

DiD Coefficient: **-0.142**

Std. Error: **0.056**

p-value: **0.011 ***

Sample Sizes

N Treatment: 572

N Control: 2,159

Ratio (C:T): **3.8 : 1**

Parallel Trends Test

p = 0.091 ✓ Assumption satisfied

✓ Pros

PT assumption holds well

Narrower control window

More comparable groups

✗ Cons

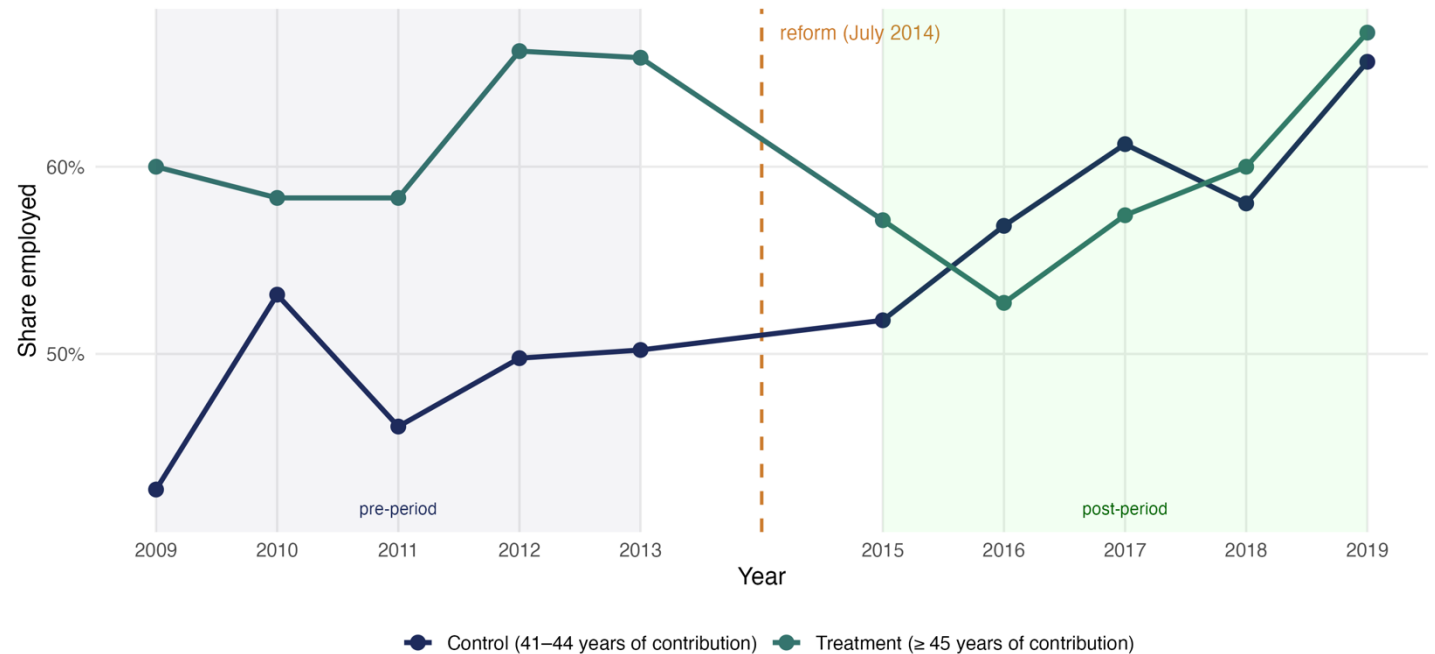
Smaller N (2,159 control)

Reduced precision

Ratio 3.8:1 still uneven

Employed: treatment- vs. controlgroup (2009–2019)

Control: 41–44 contribution years | 2014 excluded | Parallel Trends: p = 0.091



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Model 4 — Control Group: 42–44 Contribution Years



Regression Results

DiD Coefficient: **-0.130**

Std. Error: **0.058**

p-value: **0.026 ***

Sample Sizes

N Treatment: 572

N Control: **1,496**

Ratio (C:T): **2.6 : 1**

Parallel Trends Test

p = 0.097 ✓ Assumption satisfied

✓ Pros

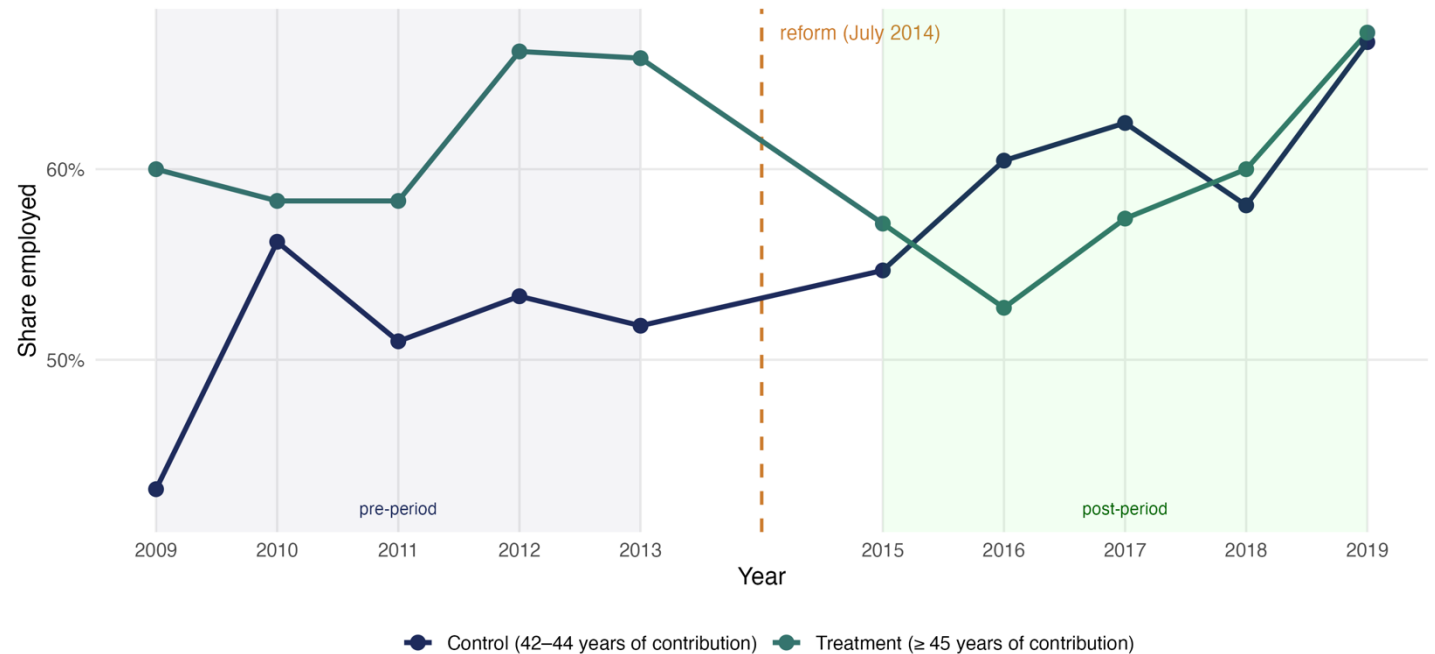
Best PT result (p=0.097)
Most comparable groups
Most credible counterfact.

✗ Cons

Smallest N (1,496 ctrl.)
Wider standard errors
Lower statistical power

Employed: treatment- vs. controlgroup (2009–2019)

Control: 42–44 contribution years | 2014 excluded | Parallel Trends: p = 0.097



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Overview — Four Control Group Specifications



$$\text{LFP} = \beta_0 + \beta_1 \cdot \text{Treat} + \beta_2 \cdot \text{Post} + \beta_3 \cdot (\text{Treat} \times \text{Post}) + \varepsilon$$

Model	Control Group	N Treat	N Control	Ratio (C:T)	DiD Coeff.	p-value	Parallel Trends p
35–44	35 – 44 contrib. years	572	5,981	10.5 : 1	–0.170	0.001 ***	0.144
40–44	40 – 44 contrib. years	572	2,866	5.0 : 1	–0.137	0.013 *	0.108
41–44	41 – 44 contrib. years	572	2,159	3.8 : 1	–0.142	0.011 *	0.091
42–44 ✓	42 – 44 contrib. years	572	1,496	2.6 : 1	–0.130	0.026 *	0.097 ✓

* p < 0.05 ** p < 0.01 *** p < 0.001 | DiD = Treat × Post | PT p > 0.05 = Parallel Trends assumption not rejected

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#	Model	Regression Equation	Fixed Effects	Sig.
Robustness — Alternative Control Groups				
1	m_35_44	$LFP = \beta_0 + \beta_1 \cdot \text{Treat} + \beta_2 \cdot \text{Post} + \beta_3 \cdot \text{DiD}$ Control: 35–44	—	✓
2	m_40_44	$LFP = \beta_0 + \beta_1 \cdot \text{Treat} + \beta_2 \cdot \text{Post} + \beta_3 \cdot \text{DiD}$ Control: 40–44	—	✓
3	m_41_44	$LFP = \beta_0 + \beta_1 \cdot \text{Treat} + \beta_2 \cdot \text{Post} + \beta_3 \cdot \text{DiD}$ Control: 41–44	—	✓
4	m_42_44	$LFP = \beta_0 + \beta_1 \cdot \text{Treat} + \beta_2 \cdot \text{Post} + \beta_3 \cdot \text{DiD}$ Control: 42–44	—	✓
Main Models (Control Group: 42–44 Contribution Years)				
5	m1	$LFP = \beta_0 + \beta_1 \cdot \text{Treat} + \beta_2 \cdot \text{Post}$	—	—
6	m2	$LFP = \beta_0 + \beta_1 \cdot \text{Treat} + \beta_2 \cdot \text{Post} + \beta_3 \cdot \text{DiD}$	—	✓
7	m3	$LFP = \beta_0 + \beta_1 \cdot \text{Treat} + \beta_2 \cdot \text{DiD}$ Year FE	Year FE	✓
8	m4	$LFP = \beta_0 + \beta_1 \cdot \text{Post} + \beta_2 \cdot \text{DiD}$ Individual FE	Individual FE	✓

✓ Interaction DID significant ($p < 0.05$)

✗ Interaction DID not significant



#	Model	Regression Equation	Fixed Effects	Sig.
RQ1 — Regional Heterogeneity: East vs. West				
9	m_ow_interact	$LFP = \beta_0 + \beta_1 \cdot \text{Treat} + \beta_2 \cdot \text{Post} + \beta_3 \cdot \text{DiD} + \beta_4 \cdot \text{East} + \beta_5 \cdot \text{DiD} \times \text{East}$	—	✗
10	m_ow_fe	$LFP = \beta_0 + \beta_1 \cdot \text{Post} + \beta_2 \cdot \text{DiD} + \beta_3 \cdot \text{East} + \beta_4 \cdot \text{DiD} \times \text{East} \mid \text{Ind. FE}$	Individual FE	✗
RQ2 — Socioeconomic Heterogeneity: Income				
11	m_inc_interact	$LFP = \beta_0 + \beta_1 \cdot \text{Treat} + \beta_2 \cdot \text{Post} + \beta_3 \cdot \text{DiD} + \beta_4 \cdot \text{HighInc} + \beta_5 \cdot \text{DiD} \times \text{HighInc}$	—	✗
12	m_inc_fe	$LFP = \beta_0 + \beta_1 \cdot \text{Post} + \beta_2 \cdot \text{DiD} + \beta_3 \cdot \text{HighInc} + \beta_4 \cdot \text{DiD} \times \text{HighInc} \mid \text{Ind. FE}$	Individual FE	✗
RQ2 — Socioeconomic Heterogeneity: Education				
13	m_edu_interact	$LFP = \beta_0 + \beta_1 \cdot \text{Treat} + \beta_2 \cdot \text{Post} + \beta_3 \cdot \text{DiD} + \beta_4 \cdot \text{HighEduc} + \beta_5 \cdot \text{DiD} \times \text{HighEduc}$	—	✗
14	m_edu_fe	$LFP = \beta_0 + \beta_1 \cdot \text{Post} + \beta_2 \cdot \text{DiD} + \beta_3 \cdot \text{HighEduc} + \beta_4 \cdot \text{DiD} \times \text{HighEduc} \mid \text{Ind. FE}$	Individual FE	✗
Dynamic DiD — Long-Run Effects				
15	m_dynamic	$LFP = \beta_0 + \beta_1 \cdot \text{Treat} + \sum \beta_t \cdot (\text{Year} \times \text{Treat}) \mid \text{Year FE} \text{ [ref: 2013]}$	Year FE	✓

✓ Interaction DID significant ($p < 0.05$)

✗ Interaction DiD not significant

Regression Results — m1 & m2



Regression Equations

Model (1): $LFP = \beta_0 + \beta_1 \cdot \text{Treat} + \beta_2 \cdot \text{Post} + \varepsilon$

Model (2): $LFP = \beta_0 + \beta_1 \cdot \text{Treat} + \beta_2 \cdot \text{Post} + \beta_3 \cdot (\text{Treat} \times \text{Post}) + \varepsilon$

Variable	(1) Without Interaction	(2) DiD Model
Constant (β_0)	0.536*** (0.024)	0.516*** (0.025)
Treatment — β_1 (≥ 45 contribution years)	0.042 (0.029)	0.110** (0.043)
Post-Period — β_2 (2015–2019)	0.060* (0.028)	0.096*** (0.032)
Treatment $\times$ Post — β_3 (DiD Estimator)	—	−0.130* (0.058)
N (Observations)	2,068	2,068
R ²	0.005	0.008

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Regression Results — m3 & m4



Regression Equations

Model (3): $LFP = \beta_1 \cdot \text{Treat} + \beta_3 \cdot \text{DiD} + \delta_t + \varepsilon$

Model (4): $LFP = \beta_2 \cdot \text{Post} + \beta_3 \cdot \text{DiD} + \alpha_i + \varepsilon$

Variable	(3) Year FE	(4) Individual FE
Treatment — β_1 (≥ 45 contribution years)	0.110 ** (0.043)	—
Post-Period — β_2 (2015–2019)	—	−0.342*** (0.057)
Fixed Effects	Year	Individual
Treatment × Post — β_3 (DiD Estimator)	−0.125* (0.058)	−0.219*** (0.050)
N (Observations)	2,068	1,674
Adj. R ² / Within R ²	0.008 / 0.005	0.499 / 0.097

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Regression Results — RQ1 Regional Heterogeneity: East vs. West



Regression Equations

OLS: $LFP = \beta_0 + \beta_1 \text{Treat} + \beta_2 \text{Post} + \beta_3 \text{DiD} + \beta_4 \text{East} + \beta_5 \text{DiD} \times \text{East} + \varepsilon$

Ind. FE: $LFP = \beta_2 \text{Post} + \beta_3 \text{DiD} + \beta_4 \text{East} + \beta_5 \text{DiD} \times \text{East} + \alpha_i + \varepsilon$

Variable	OLS	Individual FE
Treatment — β_1 (≥ 45 contr. years)	0.108** (0.043)	—
Post-Period — β_2 (2015–2019)	0.099*** (0.032)	−0.347*** (0.056)
Treatment × Post — β_3 (DiD Estimator)	−0.139** (0.061)	−0.173*** (0.052)
East — β_4	−0.032 (0.036)	—
DiD × East — β_5	0.047 (0.090)	−0.172 (0.111)
N (Observations)	2,068	1,674
Adj. R ² / Within R ²	0.009 / —	0.499 / 0.097

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Regression Results — RQ 2.1: Education



Regression Equations

OLS: $LFP = \beta_0 + \beta_1 \text{Treat} + \beta_2 \text{Post} + \beta_3 \text{DiD} + \beta_4 \text{HighEduc} + \beta_5 \text{DiD} \times \text{HighEduc} + \varepsilon$

Ind. FE: $LFP = \beta_2 \text{Post} + \beta_3 \text{DiD} + \beta_5 \text{DiD} \times \text{HighEduc} + \alpha_i + \varepsilon$

Variable	OLS	Individual FE
Treatment — β_1 (≥ 45 contr. years)	0.119** (0.044)	—
Post-Period — β_2 (2015–2019)	0.086*** (0.032)	−0.342*** (0.057)
Treatment × Post — β_3 (DiD Estimator)	−0.144** (0.063)	−0.212*** (0.058)
High Education — β_4	0.068** (0.031)	—
DiD × High Educ — β_5	0.053 (0.077)	−0.016 (0.094)
N (Observations)	2,068	1,674
Adj. R ² / Within R ²	0.014 / —	0.498 / 0.097

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Regression Results — RQ 2.2.1: Income High/Low



Regression Equations

OLS: $LFP = \beta_0 + \beta_1 \text{Treat} + \beta_2 \text{Post} + \beta_3 \text{DiD} + \beta_4 \text{HighInc} + \beta_5 \text{DiD} \times \text{HighInc} + \varepsilon$

Ind. FE: $LFP = \beta_2 \text{Post} + \beta_3 \text{DiD} + \beta_4 \text{HighInc} + \beta_5 \text{DiD} \times \text{HighInc} + \alpha_i + \varepsilon$

Variable	OLS	Individual FE
Treatment — β_1 (≥ 45 contr. years)	0.114 ** (0.042)	—
Post-Period — β_2 (2015–2019)	0.074** (0.032)	−0.346*** (0.056)
Treatment × Post — β_3 (DiD Estimator)	−0.135 (0.078)	−0.329*** (0.078)
High Income — β_4	0.204*** (0.029)	0.084** (0.032)
DiD × High Income — β_5	0.003 (0.076)	0.160 (0.085)
N (Observations)	2,068	1,674
Adj. R ² / Within R ²	0.048 / —	0.521 / 0.122

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Regression Results — RQ 2.2.2: Income Quintiles



Regression Equations

Ind. FE: $LFP = \beta_2 Post + \beta_3 DiD + \sum \beta_k (DiD \times Q_k) + \alpha_i + \varepsilon$
Income quintile interactions (Q1 = reference)

Variable

Individual FE

Post — β_2 (2015–2019)

−0.337***

(0.057)

DiD β_3 (Ref: Q1)

−0.351***

(0.099)

DiD × Q2

0.042

(0.114)

DiD × Q3

0.121

(0.123)

DiD × Q4

0.133

(0.124)

DiD × Q5 (highest quintile)

0.305**

(0.110)

N (Observations)

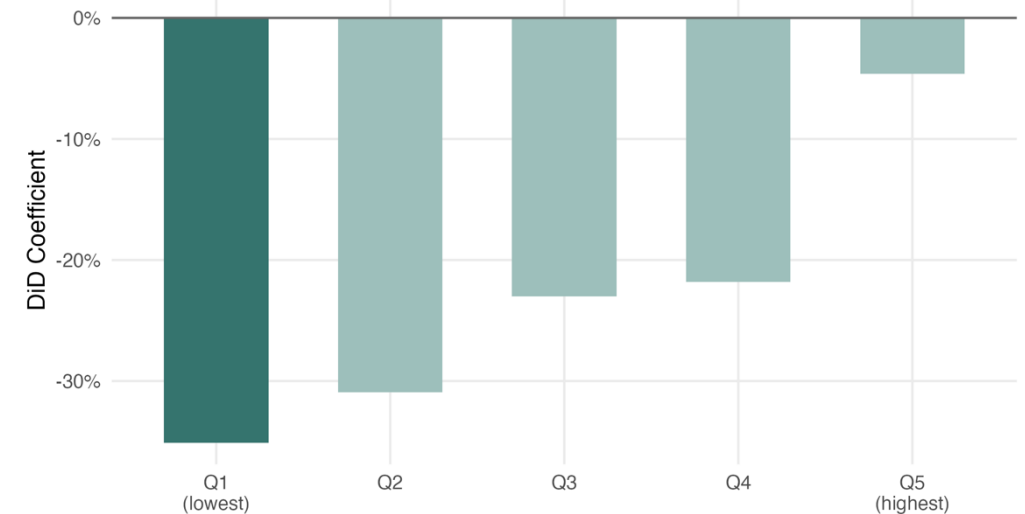
1,674

Within R²

0.103

* p < 0.05, ** p < 0.01, *** p < 0.001

Treatment Effect by Income Quintile



SOEP-Data | Age 62–65 | Control: 42–44 contribution years | Böhm & Wilhelmi 2026

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Econometric Models — Overview

Dependent Variable: LFP (1 = employed, 0 = not) | Sample: Men aged 62–65 | Clustered SE (pid)

Main Thesis

m_fe

$$\text{LFP} = \beta_0 + \beta_2 \cdot \text{Post} + \beta_3 \cdot \text{DiD} + \alpha_i + \varepsilon$$

Tests whether Rente mit 63 reduced LFP for eligible men via DiD

RQ1: East–West

m_ow_fe

$$\text{LFP} = \beta_2 \text{Post} + \beta_3 \text{DiD} + \beta_4 \text{East} + \beta_5 \text{DiD} \times \text{East} + \alpha_i + \varepsilon$$

Tests regional heterogeneity in treatment effect

RQ2a: Income

m_inc_fe

$$\text{LFP} = \beta_2 \text{Post} + \beta_3 \text{DiD} + \beta_4 \text{HighInc} + \beta_5 \text{DiD} \times \text{HighInc} + \alpha_i + \varepsilon$$

Tests differential effect by pre-reform income level

RQ2b: Education

m_educ_fe

$$\text{LFP} = \beta_2 \text{Post} + \beta_3 \text{DiD} + \beta_5 \text{DiD} \times \text{HighEduc} + \alpha_i + \varepsilon$$

Tests differential effect by education level

Robustness Check – Flexi Rente (2017)



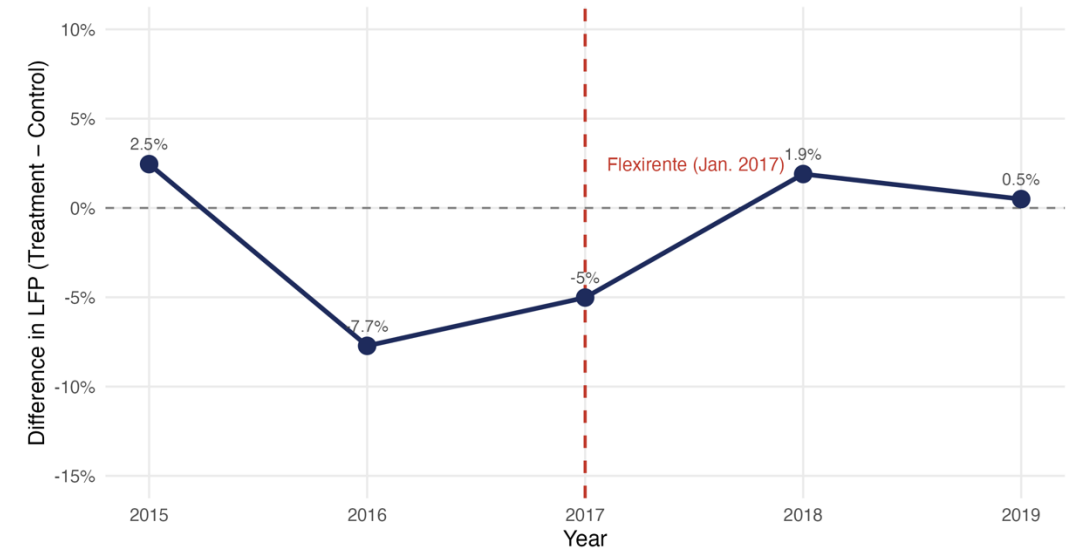
Robustness Checks

Baseline DiD, Flexi-Dummy control, and short post-period

Variable	Baseline	Flexi-Dummy	2015–16 only
Post	-0.342*** (0.057)	-0.396*** (0.057)	-0.410*** (0.064)
Treat × Post (DiD)	-0.219*** (0.050)	-0.114* (0.052)	-0.088 (0.079)
Post-2017 (Flexirente)		-0.342*** (0.045)	
Num. Obs.	1674	1674	1072
R ²	0.685	0.706	0.745
RMSE	0.278	0.268	0.251
Individual FE	Yes	Yes	Yes
Post-Period	2015–19	2015–19	2015–16

Treatment–Control Differential by Year (Post-Period)

No systematic break at Flexirente introduction (2017)



SOEP-Data | Age 62–65 | Control: 42–44 contribution years | Böhm & Wilhelmi 2026

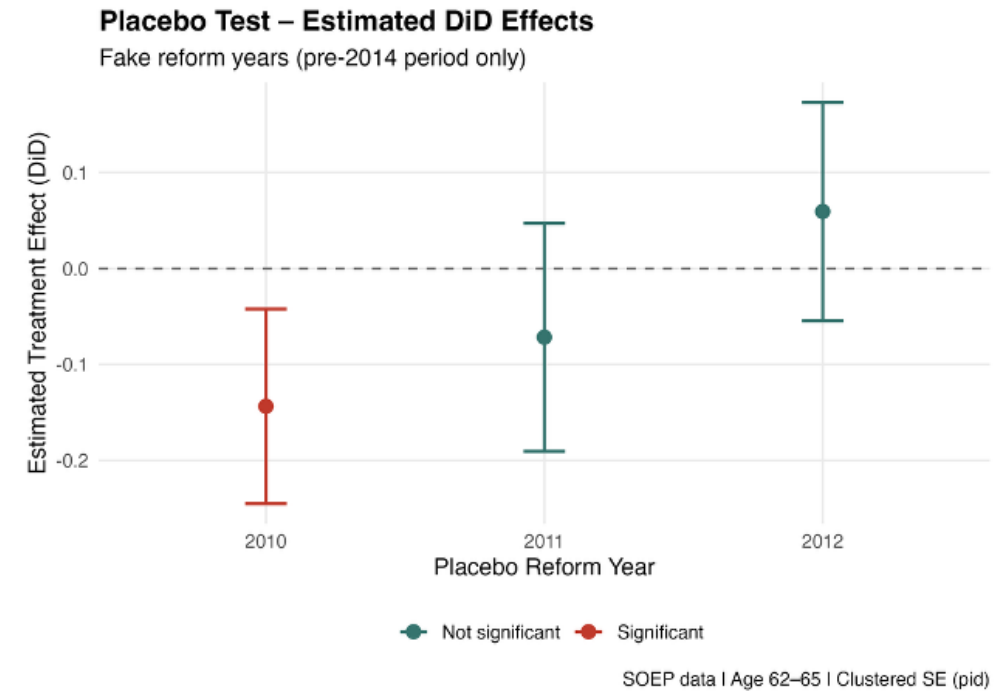
Robustness Check – Placebo Test



Placebo Design

Fake reform years (2010/2011/2012) in pre-period only (2009–2013)

Variable	Placebo 2010	Placebo 2011	Placebo 2012
Post (Placebo)	-0.175*** (0.052)	-0.216*** (0.048)	-0.233*** (0.042)
Treat × Post (Placebo DiD)	-0.144** (0.052)	-0.072 (0.061)	0.059 (0.058)
N	723	723	723
R ²	0.761	0.764	0.765



Robustness Check – Donut Test



Robustness Check – Donut Design
Excluding contribution years 44–45 | Dep. Var.: LFP | Age 62–65

Variable	Main		RQ1		RQ2 Income		RQ2 Education	
	Main (Baseline)	Main (Donut)	RQ1 (Baseline)	RQ1 (Donut)	RQ2 Inc (Baseline)	RQ2 Inc (Donut)	RQ2 Edu (Baseline)	RQ2 Edu (Donut)
Post	−0.342*** (0.057)	−0.339*** (0.077)	−0.347*** (0.056)	−0.345*** (0.076)	−0.346*** (0.056)	−0.342*** (0.077)	−0.342*** (0.057)	−0.343*** (0.076)
Treat × Post	−0.219*** (0.050)	−0.194** (0.073)	−0.173*** (0.052)	−0.116 (0.075)	−0.329*** (0.078)	−0.310** (0.100)	−0.212*** (0.058)	−0.142 (0.080)
Treat × Post × East			−0.172 (0.111)	−0.259 (0.135)				
Treat × Post × High Income					0.160 (0.085)	0.170 (0.102)		
Treat × Post × High Education							−0.016 (0.094)	−0.114 (0.121)
Num. Obs.	1674	1232	1674	1232	1674	1232	1674	1232
R ²	0.685	0.718	0.686	0.720	0.686	0.719	0.685	0.718
RMSE	0.278	0.265	0.278	0.264	0.278	0.264	0.278	0.264
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Donut	No	Yes	No	Yes	No	Yes	No	Yes

* p < 0.05, ** p < 0.01, *** p < 0.001
Clustered SE auf pid-Ebene | Donut: contrib. years 44–45 excluded

Robustness Check – Donut Test



Robustness Check – Donut Design

Excluding contribution years 44–45 | Dep. Var.: LFP | Age 62–65

Main

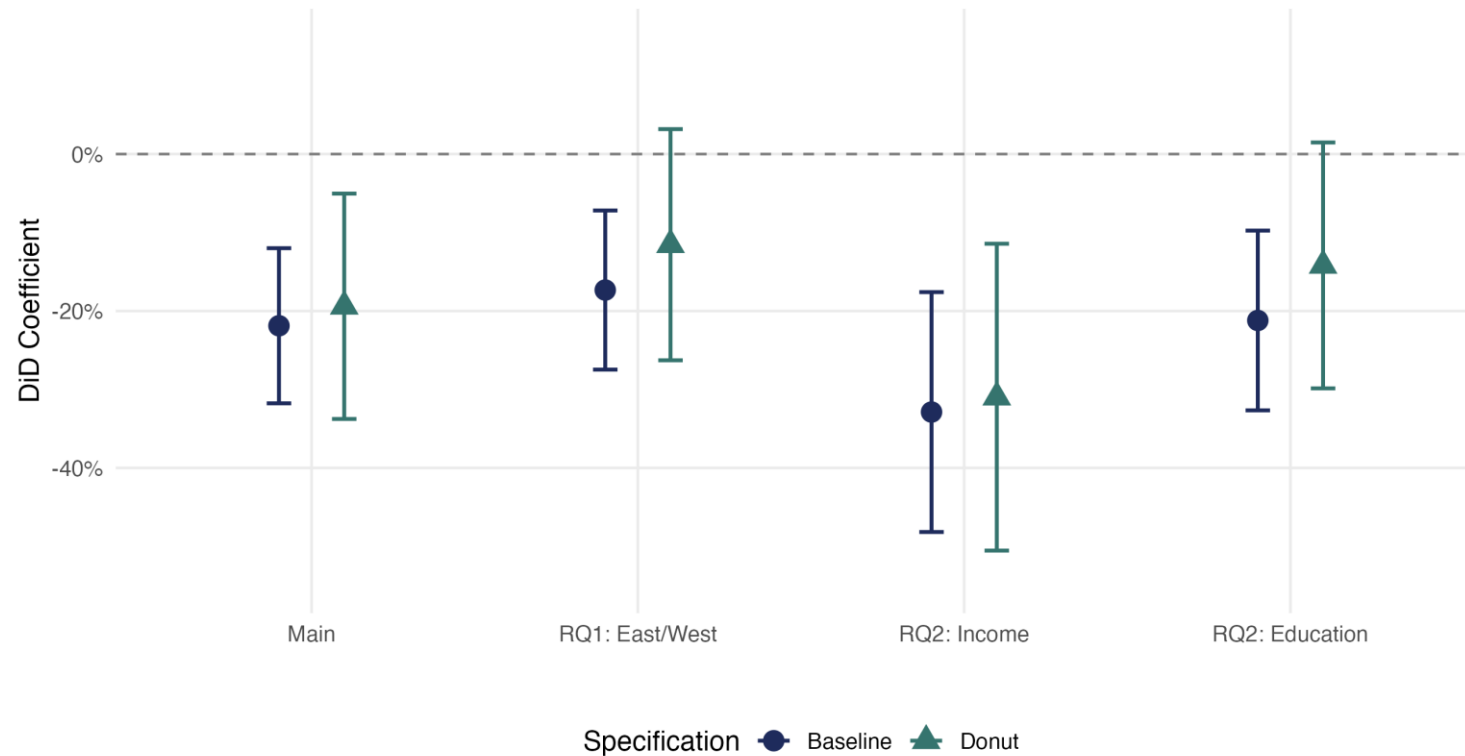
RQ1

RQ2: Income

RQ2: Education

Robustness Check – Donut Design

DiD coefficient (Treat × Post) | Baseline vs. excluding contrib. years 44–45



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